

3.4

- (a) For the Fourier transform of $x[n]$ to exist, the z -transform of $x[n]$ must have an ROC which includes the unit circle, therefore, $|\frac{1}{3}| < |z| < |2|$.
 Since this ROC lies outside $\frac{1}{3}$, this pole contributes a right-sided sequence. Since the ROC lies inside 2 and 3, these poles contribute left-sided sequences. The overall $x[n]$ is therefore two-sided.
- (b) Two-sided sequences have ROC's which look like washers. There are two possibilities. The ROC's corresponding to these are: $|\frac{1}{3}| < |z| < |2|$ and $|2| < |z| < |3|$.
- (c) The ROC must be a connected region. For stability, the ROC must contain the unit circle. For causality the ROC must be outside the outermost pole. These conditions cannot be met by any of the possible ROC's of this pole-zero plot.

3.6

- (a) one pole $\rightarrow$ inspection

$$x(z) = \frac{1}{1 + \frac{1}{2}z^{-1}} = \frac{1}{1 - (-\frac{1}{2})z^{-1}}, a = -\frac{1}{2}$$

$$\therefore |z| > \frac{1}{2}, \text{right-sided}$$

$$\therefore x[n] = (-\frac{1}{2})^n u[n]$$

- (b) one pole $\rightarrow$ inspection

$$x(z) = \frac{1}{1 + \frac{1}{2}z^{-1}} = \frac{1}{1 - (-\frac{1}{2})z^{-1}}, a = -\frac{1}{2}$$

$$\therefore |z| < \frac{1}{2}, \text{left-sided}$$

$$\therefore x[n] = -(-\frac{1}{2})^n u[-n-1]$$

- (c)

$$x(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 + \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}} = \frac{1 - \frac{1}{2}z^{-1}}{\left(1 + \frac{1}{4}z^{-1}\right)\left(1 + \frac{1}{2}z^{-1}\right)} = \frac{-3}{1 + \frac{1}{4}z^{-1}} + \frac{4}{1 + \frac{1}{2}z^{-1}}$$

$$\therefore |z| > \frac{1}{2}, \text{right-sided}$$

$$\therefore x[n] = (-3)\left(-\frac{1}{4}\right)^n u[n] + 4\left(-\frac{1}{2}\right)^n u[n]$$

$$= \left[(-3)\left(-\frac{1}{4}\right)^n + 4\left(-\frac{1}{2}\right)^n\right] u[n]$$

(d)

$$X(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{1}{4}z^{-2}} \quad |z| > \frac{1}{2}$$

Partial Fractions:

$$\begin{aligned} X(z) &= \frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{1}{4}z^{-2}} = \frac{1}{1 + \frac{1}{2}z^{-1}} \quad |z| > \frac{1}{2} \\ x[n] &= \left(-\frac{1}{2}\right)^n u[n] \end{aligned}$$

(e)

$$X(z) = \frac{1 - az^{-1}}{z^{-1} - a} \quad |z| > |a^{-1}|$$

Partial Fractions:

$$\begin{aligned} X(z) &= -a - \frac{a^{-1}(1 - a^2)}{1 - a^{-1}z^{-1}} \quad |z| > |a^{-1}| \\ x[n] &= -a\delta[n] - (1 - a^2)a^{-(n+1)}u[n] \end{aligned}$$

3.9

$$H(z) = \frac{1 + z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{4}z^{-1})} = \frac{2}{(1 - \frac{1}{2}z^{-1})} - \frac{1}{(1 + \frac{1}{4}z^{-1})}$$

(a) $h[n]$ causal $\Rightarrow$ ROC outside $|z| = \frac{1}{2} \Rightarrow |z| > \frac{1}{2}$.

(b) ROC includes $|z| = 1 \Rightarrow$ stable.

(c)

$$\begin{aligned} y[n] &= -\frac{1}{3} \left(-\frac{1}{4}\right)^n u[n] - \frac{4}{3} (2)^n u[-n-1] \\ Y(z) &= \frac{-\frac{1}{3}}{1 + \frac{1}{4}z^{-1}} + \frac{\frac{4}{3}}{1 - 2z^{-1}} \\ &= \frac{1 + z^{-1}}{(1 + \frac{1}{4}z^{-1})(1 - 2z^{-1})} \quad \frac{1}{4} < |z| < 2 \\ X(z) &= \frac{Y(z)}{H(z)} = \frac{(1 - \frac{1}{2}z^{-1})}{(1 - 2z^{-1})} \quad |z| < 2 \\ x[n] &= -(2)^n u[-n-1] + \frac{1}{2} (2)^{n-1} u[-n] \end{aligned}$$

(d)

$$h[n] = 2 \left(\frac{1}{2}\right)^n u[n] - \left(-\frac{1}{4}\right)^n u[n]$$

3.23

(a)

$$\begin{aligned}
 H(z) &= \frac{1 - \frac{1}{2}z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})} \\
 &= -4 + \frac{5 + \frac{7}{2}z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}} \\
 &= -4 - \frac{2}{1 - \frac{1}{2}z^{-1}} + \frac{7}{1 - \frac{1}{4}z^{-1}} \\
 h[n] &= -4\delta[n] - 2\left(\frac{1}{2}\right)^n u[n] + 7\left(\frac{1}{4}\right)^n u[n]
 \end{aligned}$$

(b)

$$\begin{aligned}
 H(z) = \frac{Y(z)}{X(z)} &= \frac{1 - \frac{1}{2}z^{-2}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}, \quad Y(z) - \frac{3}{4}z^{-1}Y(z) + \frac{1}{8}z^{-2}Y(z) = X(z) - \frac{1}{2}z^{-2}X(z), \\
 y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] &= x[n] - \frac{1}{2}x[n-2]
 \end{aligned}$$

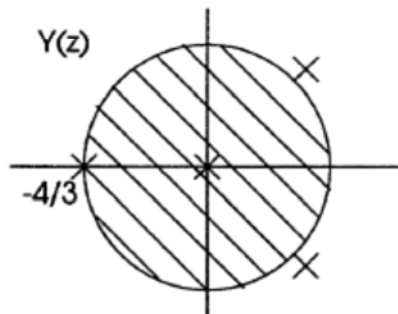
3.36

From the pole-zero diagram

$$\begin{aligned}
 X(z) &= \frac{z}{(z^2 - z + \frac{1}{2})(z + \frac{3}{4})} \quad |z| > \frac{3}{4} \\
 y[n] &= x[-n+3] = x[-(n-3)] \\
 \Rightarrow Y(z) &= z^{-3}X(z^{-1}) = \frac{z^{-3}z^{-1}}{(z^{-2} - z^{-1} + \frac{1}{2})(z^{-1} + \frac{3}{4})} \\
 &= \frac{8/3}{z(2 - 2z + z^2)(\frac{4}{3} + z)}
 \end{aligned}$$

Poles at $0, -\frac{4}{3}, 1 \pm j$, zeros at ∞

$x[n]$ causal $\Rightarrow x[-n+3]$ is left-sided $\Rightarrow$ ROC is $0 < |z| < 4/3$.



~Matlab problem:

Matlab code: HW2b.m

```
num=[2 16 44 56 32];
```

```
den=[3 3 -15 18 -12];
```

```
[z,p,k]=tf2zp(num,den);
```

```
disp('zrors');disp(z')
```

```
disp('poles');disp(p')
```

```
zplane(num,den);
```

```
[r,p,k] = residuez(num,den);
```

```
disp('Residues');disp(r')
```

```
disp('Poles');disp(p')
```

```
disp('Constants');disp(k)
```

```
>> HW2b
```

```
zrors
```

```
-4.0000 + 0.0000i  -2.0000 + 0.0000i  -1.0000 - 1.0000i  -1.0000 + 1.0000i
```

```
poles
```

```
-3.2361 + 0.0000i   1.2361 + 0.0000i   0.5000 - 0.8660i   0.5000 + 0.8660i
```

```
Residues
```

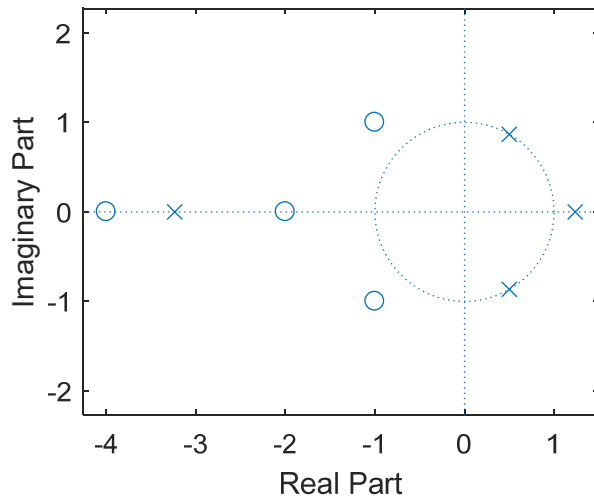
```
-0.0177 + 0.0000i   9.4914 + 0.0000i  -3.0702 - 2.3398i  -3.0702 + 2.3398i
```

```
Poles
```

```
-3.2361 + 0.0000i   1.2361 + 0.0000i   0.5000 - 0.8660i   0.5000 + 0.8660i
```

```
Constants
```

```
-2.6667
```



$$G(z) = -2.6667 + \frac{-0.0177}{1 + 3.2361z^{-1}} + \frac{9.4914}{1 - 1.236z^{-1}} + \frac{-3.0702 - 2.3398i}{1 - (0.5000 - 0.8660i)z^{-1}} + \frac{-3.0702 + 2.3398i}{1 - (0.5000 + 0.8660i)z^{-1}}$$